



CHALLENGE 01 / 20

THE MIDNIGHT LEDGER TRANSFER

FINANCIAL TRANSACTION RECONSTRUCTION & DIGITAL ATTRIBUTION

NATIONAL LEVEL HACKATHON • 2026

POWERED BY SYNTHOQUEST PVT LTD

Student Project Development Challenge • No organizer-provided evidence package.

1 | PROBLEM BACKGROUND

At 00:07 on a Sunday, the treasury of a state apex cooperative bank began moving funds through RTGS, NEFT, UPI collection rails and internal ledger adjustments. By 01:41, ₹312 crore had passed through 214 accounts at 37 institutions, split, merged and split again so that no single chain could be followed. Reversal windows closed before anyone was awake.

Every instruction carried a valid treasury credential, a valid hardware token and a valid maker-checker approval. Three operators say they were asleep. The core banking system, payment gateway, interbank messaging interface, VPN concentrator, privileged access vault and mobile channel each logged a different slice of the night, with different identifiers, clock offsets and levels of detail. Two of these systems were reconfigured while the incident was running.

Some transfers were genuine overnight settlements. Some were automated treasury sweeps. Some were injected. A few look like all three. The regulator, the police cyber cell and the bank's insurer each need a different answer, and none will accept a probability without an explanation.

2 | PROBLEM STATEMENT

Develop a **Transaction Authorship Attribution System** that determines, for every step in a multi-institution fund-movement graph, whether it was executed by an authorised human, a legitimate automated process or an impersonator, even when credentials, tokens and maker-checker approvals were genuine.

The system must separate attacker-controlled steps from legitimate steps that merely share the same path, and state which approvals were obtained by deception, which by compromise and which were honestly given.

Hard constraints

- Records disagree on identifiers, clocks and detail, and some logging was altered during the incident.
- The fund movement is a graph with hundreds of accounts, splits, merges and layering, not a single chain.
- Injected activity imitates the normal batch pattern, and the adversary has learned that pattern.
- Stated confidence must be calibrated: it must match measured accuracy against the team's own ground truth.
- The same analysis must yield differently scoped findings for a regulator, a police unit and an insurer without changing the conclusions.

No evidence package or dataset is provided. Teams decide what to build, what data they need, how to simulate the environment and how to validate the result. No tools or architecture are prescribed.

3 | MANDATORY FEATURES

Teams must develop a working software solution. The system must provide the following features. How each is achieved is for the team to decide.

01	Fund-Flow Graph Reconstruction	Ability to rebuild a layered multi-institution movement of funds from partial records.
02	Cross-System Identity Resolution	Ability to match one transfer across systems that name and time it differently.
03	Actor-Type Attribution	Ability to attribute each step to human, automation or impersonator.
04	Approval Integrity Analysis	Ability to judge whether a genuine approval was honestly given.
05	Automation Mimicry Detection	Ability to expose injected activity that copies legitimate batch behaviour.
06	Tampered-Log Handling	Ability to reason when logging was changed during the incident.
07	Calibrated Confidence	Ability to produce confidence that matches measured accuracy.
08	Audience-Specific Findings	Ability to produce consistent findings scoped for different authorities.

The system must clearly distinguish between:

- AUTHORISED
- AUTOMATED
- IMPERSONATED
- INDETERMINATE
- BYPASSED

An attribution that cannot be supported by evidence must not be presented as established.

Build, don't search. This is a development challenge. Teams must create the system and the controlled environment that proves it works. Submitting an analysis of a given dataset is not a solution.

4 | EXPECTED OUTPUT

Live demonstration. The team must show that its system can handle a complex controlled scenario created by the team. The demonstration must show:

01	Teams' own simulated bank environment with layered fund movement
02	Reconstruction of the full fund-flow graph
03	Attribution of every step by actor type
04	Detection of automation mimicry
05	Operation with altered and conflicting logs
06	Calibration of confidence against ground truth
07	Three audience-specific findings from one analysis

The team must be able to explain why the system reached each major conclusion.

Validation standard (every team)

- Build a scenario generator with known ground truth and evaluate on at least ten distinct scenarios, including ones with deliberately misleading evidence.
- At the demonstration, judges change scenario parameters on the spot. The system must adapt without being rebuilt.
- Include an adversary variant that actively tries to defeat the system, and report how the system held up.
- Stated confidence must match measured accuracy. Overconfident or unexplained conclusions lose marks.
- No real systems, real personal data or functional malware may be used.

Submission. Each team must submit:

01	Working Prototype	A functional implementation of the proposed solution.
02	Source Code	Complete source code required to reproduce the project.
03	Technical Architecture	Architecture and component description.
04	Validation Environment	A controlled environment created by the team to demonstrate the solution.
05	Technical Documentation	Implementation, design and testing documentation.
06	Forensic Investigation Report	A professional report generated from the demonstrated investigation.
07	Demonstration	Live technical demonstration to the judging panel.
08	Limitations	Known limitations and situations where the system cannot establish a reliable conclusion.